



# Biliary Tract Disease: Cholangiopathies, Gallstones & Tumors

Pathology GIT Block · MEDucation by Mattin Majid

## BEFORE YOU READ — ANSWER FROM MEMORY

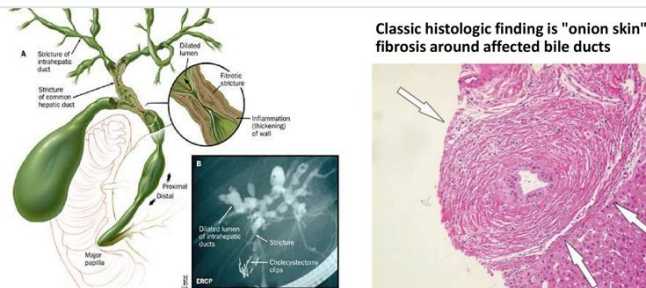
1. What distinguishes PBC from PSC by duct distribution, antibody, sex, and histology?
2. What drives cholesterol gallstone formation, and how do black and brown pigment stones differ in cause?
3. What separates acute calculous from acute acalculous cholecystitis, and which carries higher mortality?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

## 1 Autoimmune cholangiopathies

Two disorders: **primary biliary cholangitis (PBC)** and **primary sclerosing cholangitis (PSC)**.

| Feature                  | PBC                                                                                                                                                                   | PSC                                                                                                                                   |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>Lesion</b>            | Non-suppurative, inflammatory destruction of <b>small/medium intrahepatic (interlobular)</b> bile ducts; large intrahepatic ducts and extrahepatic tree <b>spared</b> | Inflammation + obliterative fibrosis of <b>intra- and extrahepatic</b> bile ducts → dilation of preserved segments ("beading" on MRI) |
| <b>Age / sex</b>         | Middle-age <b>women</b> ; peak 40–50y ( <b>90% female</b> )                                                                                                           | 3rd–5th decades; <b>2:1 male</b> predominance                                                                                         |
| <b>Key antibody</b>      | <b>Antimitochondrial antibody (AMA)</b> in 90–95%                                                                                                                     | <b>Anti-smooth muscle Ab</b> (~75%), ANA, ANCA; AMA usually negative                                                                  |
| <b>Association</b>       | Sjögren syndrome (70%), scleroderma, thyroid disease                                                                                                                  | <b>Inflammatory bowel disease</b> (mostly UC) in ~70%                                                                                 |
| <b>Classic histology</b> | Florid duct lesions, loss of small ducts                                                                                                                              | <b>"Onion-skin" concentric fibrosis</b> around affected bile ducts                                                                    |
| <b>Course</b>            | Progressive → cirrhosis + profound cholestasis, or prominent portal hypertension (both now rare with therapy)                                                         | Unpredictable — may progress to <b>cholangiocarcinoma</b>                                                                             |

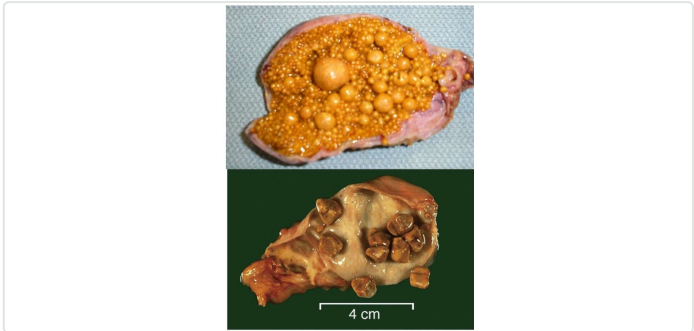


PSC: strictures and beading of the biliary tree (gross/ERCP) and the classic "onion-skin" periductal fibrosis on histology

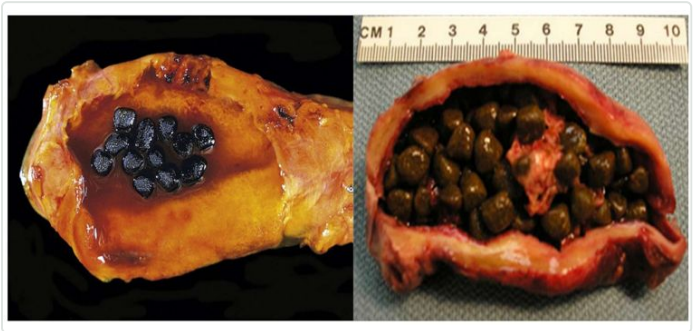
## 2 Gallstones

|                                   |                                                                                                                                                                                                                                                 |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Two main types</b>             | <b>Cholesterol stones</b> (crystalline cholesterol monohydrate) · <b>pigment stones</b> (calcium bilirubinate salts)                                                                                                                            |
| <b>Cholesterol pathogenesis</b>   | Cholesterol is kept soluble by bile salts/lecithin; when cholesterol <b>supersaturates</b> bile, it crystallizes out. Enhanced by <b>gallbladder hypomotility/stasis</b> (crystal nucleation) and <b>mucus hypersecretion</b> (traps crystals). |
| <b>Cholesterol stone subtypes</b> | <b>Pure</b> (~10%, ≥90% cholesterol, pale yellow, often multiple, usually radiolucent) vs <b>mixed</b> (more common, ≥60% cholesterol + calcium salts, <b>radiopaque</b> )                                                                      |
| <b>Pigment stone pathogenesis</b> | Form when bile has high <b>unconjugated bilirubin</b> — from chronic hemolysis or biliary infection                                                                                                                                             |
| <b>Pigment stone subtypes</b>     | <b>Black</b> (sterile bile, intravascular hemolysis, small/numerous/fragile, 50–75% radiopaque) vs <b>brown</b> (infected ducts, few, soft/greasy, usually radiolucent)                                                                         |

| Risk factors              | Detail                                                                                                                                                                              |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Cholesterol stones</b> | Advancing age · female sex hormones (female sex, OCPs, pregnancy) · obesity/insulin resistance · rapid weight loss · gallbladder stasis · inborn bile-acid disorders · dyslipidemia |
| <b>Pigment stones</b>     | Chronic hemolysis (sickle cell, hereditary spherocytosis) · biliary infection · GI disorders (Crohn disease, ileal resection/bypass, CF with pancreatic insufficiency)              |



Cholesterol gallstones: pure (top, pale yellow/round) and mixed (bottom, faceted, calcium-containing)



Pigment gallstones: black (left, sterile bile/hemolysis) and brown (right, infected ducts)

Outcomes and complications of gallstones

Biliary colic · cholecystitis · mucocele · empyema · fistula · perforation · **gallbladder carcinoma** — or, if stones migrate out of the gallbladder: common hepatic duct obstruction · choledocholithiasis · cholangitis · intrahepatic stone · **pancreatitis** · gallstone ileus.

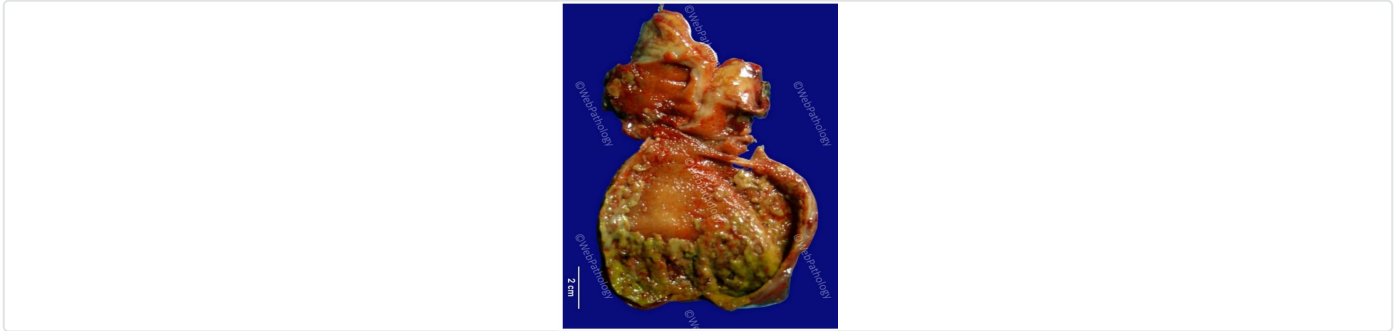
**Outcomes and complications of GS**

1. Biliary colic.
2. Cholecystitis.
3. Mucocele.
4. Empyema.
5. Fistula.
6. Perforation.
7. Gallbladder carcinoma.
8. Common hepatic duct obstruction.
9. Choledocholithiasis.
10. Cholangitis.
11. Intrahepatic stone.
12. Pancreatitis.
13. Gall stone ileus.

Complications mapped to site: in the gallbladder (1-7) vs after stones migrate into the ducts (8-13)

3 Cholecystitis

| <b>Forms</b>                 | <b>Acute</b> (calculous or acalculous) · <b>chronic</b> · acute-on-chronic                                                                                                                                                                   |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Epidemiology</b>          | Almost always occurs with gallstones; distribution parallels that of gallstones                                                                                                                                                              |
| Type                         | Key features                                                                                                                                                                                                                                 |
| <b>Acute calculous (90%)</b> | Gallstone obstructs the <b>gallbladder neck/cystic duct</b> . RUQ pain, biliary colic, nausea, vomiting, low-grade fever, leukocytosis. Complications: gangrene, perforation, peritonitis, fistula, gallstone ileus.                         |
| <b>Acute acalculous</b>      | From gallbladder <b>stasis + ischemia</b> in seriously ill patients — major surgery, severe trauma/burns, sepsis. <b>Higher mortality</b> (more gangrene/perforation).                                                                       |
| <b>Chronic</b>               | May follow repeat acute attacks, or arise without them; almost always with gallstones (rarely acalculous). Bile supersaturation predisposes to both inflammation and stone formation. <b>E. coli/enterococci</b> cultured from bile in ~1/3. |

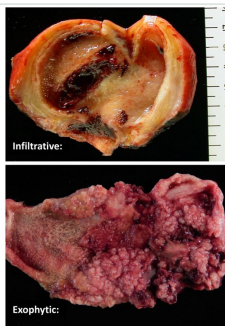


Acute calculous cholecystitis: mucosal erythema, edema, congestion to ulceration/necrosis, with patchy greenish-yellow purulent exudate

## 4 Tumors of the gallbladder and bile ducts

### Carcinoma of the gallbladder

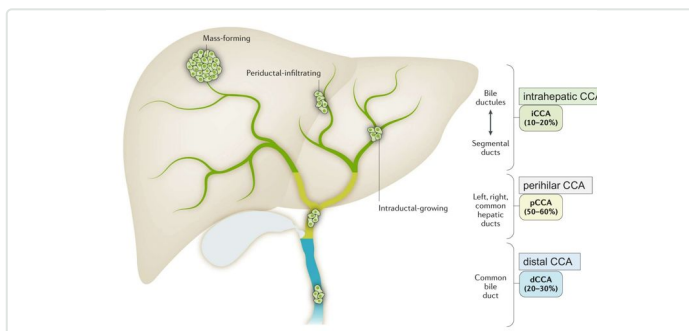
|                       |                                                                                                                                             |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>What</b>           | <b>Most common malignancy of the extrahepatic biliary tract</b>                                                                             |
| <b>Risk factors</b>   | <b>Gallstones</b> (present in 95% of cases, via chronic inflammation) · carcinogenic bile-acid derivatives · primary sclerosing cholangitis |
| <b>Clinical</b>       | Usually found incidentally at surgery for an unrelated condition; symptoms insidious, mimic cholelithiasis                                  |
| <b>Gross patterns</b> | <b>Infiltrative</b> (diffuse wall thickening/induration) or <b>exophytic</b> (irregular mass into the lumen)                                |
| <b>Microscopic</b>    | Mostly <b>adenocarcinoma</b> ; ~5% squamous cell carcinoma                                                                                  |



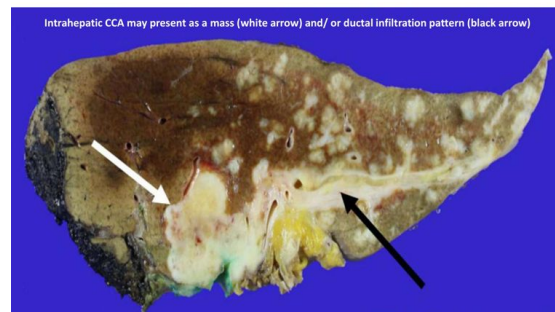
Gallbladder carcinoma: infiltrative pattern (top, diffuse wall thickening) vs exophytic pattern (bottom, mass into the lumen)

### Cholangiocarcinoma (CCA)

|                       |                                                                                                                                                                                                                      |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>What</b>           | Carcinoma of <b>intrahepatic and extrahepatic bile ducts</b> ; histologically <b>adenocarcinoma</b>                                                                                                                  |
| <b>Risk factors</b>   | Conditions of chronic inflammation: <b>liver fluke infestation</b> · <b>primary sclerosing cholangitis</b> · intrahepatic gallstones · hepatitis B/C · NAFLD                                                         |
| <b>Classification</b> | By location — <b>intrahepatic CCA</b> (10–20%, bile ductules/segmental ducts) · <b>perihilar CCA</b> (50–60%, left/right/common hepatic ducts — the most common site) · <b>distal CCA</b> (20–30%, common bile duct) |



CCA classified by location: intrahepatic (mass-forming, periductal-infiltrating, or intraductal-growing), perihilar, and distal



Intrahepatic CCA: may present as a mass (white arrow) and/or ductal infiltration (black arrow)

### P Must-remember pearls

- **PBC**: AMA-positive (90–95%), middle-age women, small/medium intrahepatic ducts. **PSC**: ANCA/ASMA, young men, intra- and extrahepatic ducts with "onion-skin" fibrosis, linked to UC, risk of cholangiocarcinoma.
- Gallstones are **cholesterol** (supersaturation + stasis + mucus hypersecretion) or **pigment** (black = hemolysis/sterile bile; brown = infected ducts).
- Acute **calculous** cholecystitis (90%) = cystic duct obstruction by a stone; acute **acalculous** cholecystitis = stasis + ischemia in critically ill patients, with higher mortality.
- **Gallbladder carcinoma** is the commonest extrahepatic biliary malignancy, strongly gallstone-associated (95%), usually an incidental surgical finding.
- **Cholangiocarcinoma** arises from chronic biliary inflammation (flukes, PSC, hepatolithiasis, viral hepatitis); **perihilar** is the most common location (50–60%).

## Q Test yourself

**1. A 45-year-old woman has pruritus and fatigue. Antimitochondrial antibodies are strongly positive, and biopsy shows destruction of small intrahepatic bile ducts with sparing of the extrahepatic tree. This is most consistent with:**

- A. Primary biliary cholangitis
- B. Primary sclerosing cholangitis
- C. Acute cholecystitis
- D. Cholangiocarcinoma
- E. Chronic cholecystitis

**2. A 30-year-old man with ulcerative colitis develops cholestasis. ERCP shows a beaded pattern of intra- and extrahepatic bile ducts, and biopsy shows concentric "onion-skin" periductal fibrosis. The main long-term malignant risk is:**

- A. Hepatocellular carcinoma
- B. Gallbladder carcinoma
- C. Pancreatic adenocarcinoma
- D. Cholangiocarcinoma
- E. Colonic adenocarcinoma

**3. A patient with chronic hemolytic anemia develops small, black, fragile gallstones. These are best classified as:**

- A. Pure cholesterol stones
- B. Mixed cholesterol stones
- C. Pigment stones
- D. Calcium phosphate stones
- E. Struvite stones

**4. A critically ill patient in the ICU after major trauma develops right upper quadrant pain and fever. Imaging shows gallbladder wall thickening without stones. This is most consistent with:**

- A. Acute calculous cholecystitis
- B. Chronic cholecystitis
- C. Primary biliary cholangitis
- D. Choledocholithiasis
- E. Acute acalculous cholecystitis

**5. A patient undergoing cholecystectomy for gallstones is incidentally found to have a gallbladder mass. Biopsy shows adenocarcinoma. The strongest known risk factor for this tumor is:**

- A. Hepatitis B infection
- B. Gallstones
- C. Smoking
- D. Alcohol use
- E. Obesity alone

### ANSWERS — COVER UNTIL YOU'VE COMMITTED

- 1 • A** — PBC is an AMA-positive (90-95%) autoimmune disease of middle-aged women, destroying small/medium intrahepatic bile ducts while sparing large intrahepatic and extrahepatic ducts.
- 2 • D** — PSC, strongly associated with ulcerative colitis and showing "onion-skin" fibrosis, may progress to cholangiocarcinoma.
- 3 • C** — Black pigment stones form in sterile gallbladder bile and are associated with intravascular hemolysis (chronic hemolytic conditions).
- 4 • E** — Acute acalculous cholecystitis results from gallbladder stasis and ischemia, typically in seriously ill patients (major surgery, trauma, burns, sepsis), and carries a higher mortality than the calculous form.
- 5 • B** — Gallstones are present in about 95% of gallbladder carcinoma cases, likely via chronic inflammation, making them the strongest associated risk factor.

### SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam

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Study alongside the lecture, not instead of it.